DEFENCE JOURNEY

NDA Chemistry Notes

Key facts, reactions and everyday chemistry for the NDA General Science paper

NDA | CHEMISTRY

Chemistry questions in NDA are mostly from the NCERT level: atomic structure, the periodic table, acids and bases, metals, carbon compounds and everyday chemistry. Learn the facts and equations below.

1. Matter and Atomic Structure

- Matter is anything that has mass and occupies space. States: solid, liquid, gas, plasma (and Bose-Einstein condensate).
- **Element** is made of one type of atom. **Compound** has fixed ratio of two or more elements chemically combined. **Mixture** can be separated by physical methods.
- Atom = nucleus (protons + neutrons) + electrons. Proton: +1 charge. Electron: −1 charge. Neutron: no charge.
- **Atomic number** = number of protons. **Mass number** = protons + neutrons.
- **Isotopes:** same atomic number, different mass number (C-12 and C-14). **Isobars:** same mass number, different atomic number. **Isotones:** same number of neutrons.
- Discoveries: electron – J. J. Thomson; proton – Rutherford (Goldstein discovered anode rays); neutron – James Chadwick; nucleus – Rutherford.
- Bohr model: electrons move in fixed orbits. Maximum electrons in a shell = $2n^2$ (K = 2, L = 8, M = 18).
- Valency is the combining capacity of an atom, decided by the electrons in the outermost shell.

2. Periodic Table

- Modern periodic law: properties of elements are a periodic function of their **atomic number**. (Mendeleev used atomic mass.)
- 18 groups and 7 periods. Group 1: alkali metals. Group 2: alkaline earth metals. Group 17: halogens. Group 18: noble gases.
- Across a period (left to right): atomic size decreases, ionisation energy and electronegativity increase, metallic character decreases.
- Down a group: atomic size increases, ionisation energy decreases, metallic character increases.
- Most electronegative element: **fluorine**. Largest atom in the table: caesium (stable, naturally occurring). Lightest element: hydrogen.
- Noble gases (He, Ne, Ar, Kr, Xe, Rn) are almost non-reactive because their outer shell is full.

3. Chemical Bonding

Type	How formed	Properties	Example
Ionic	transfer of electrons (metal + non-metal)	high melting point, conduct when molten or dissolved	NaCl, MgO
Covalent	sharing of electrons (non-metal + non-metal)	low melting point, poor conductors	H_2O, CH_4, CO_2
Metallic	sea of free electrons	good conductors, malleable, ductile	Cu, Fe

- Water has a bent shape. Methane is tetrahedral. Carbon dioxide is linear. Ammonia is pyramidal.
- Hydrogen bonding is responsible for the high boiling point of water.

4. Acids, Bases and Salts

- Acids turn blue litmus red, taste sour, pH below 7. Bases turn red litmus blue, feel soapy, pH above 7. Neutral pH = 7.
- **Arrhenius:** acid gives H^+, base gives OH^- in water. **Bronsted-Lowry:** acid donates a proton, base accepts a proton.
- Strong acids: HCl, H_2SO_4, HNO_3. Weak acids: acetic acid (vinegar), citric acid, carbonic acid. Strong bases: NaOH, KOH.
- Neutralisation: acid + base → salt + water.
- Natural sources: lemon and orange (citric acid), vinegar (acetic acid), tamarind (tartaric acid), curd (lactic acid), ant sting and nettle (formic acid), tomato (oxalic acid), apple (malic acid).
- **pH of common substances:** gastric juice about 1.5 to 2, lemon juice 2.2, pure water 7, blood 7.4, baking soda solution 8.3, milk of magnesia 10.
- Antacids (milk of magnesia, baking soda) neutralise excess stomach acid. Tooth decay begins when pH of the mouth falls below 5.5.
- Common salt NaCl; baking soda $NaHCO_3$; washing soda $Na_2CO_3 \cdot 10H_2O$; bleaching powder $CaOCl_2$; plaster of Paris $CaSO_4 \cdot \frac{1}{2}H_2O$; quicklime CaO; slaked lime $Ca(OH)_2$; limestone $CaCO_3$.

5. Metals and Non-metals

- Metals are lustrous, malleable, ductile and good conductors. Mercury is the only metal that is liquid at room temperature. Gallium and caesium melt close to room temperature.
- Non-metals are generally brittle and poor conductors. **Graphite** (a non-metal) conducts electricity. **Diamond** is the hardest natural substance.
- Reactivity series (high to low): K, Na, Ca, Mg, Al, Zn, Fe, Pb, H, Cu, Hg, Ag, Au.
- Gold, silver and platinum are least reactive. Potassium and sodium are stored in kerosene.
- **Rusting** needs both air (oxygen) and moisture. It is prevented by painting, galvanisation (zinc coating), and alloying.
- **Alloys:** Brass (Cu + Zn), Bronze (Cu + Sn), Steel (Fe + C), Stainless steel (Fe + Cr + Ni), Solder (Pb + Sn), Duralumin (Al + Cu + Mg + Mn).
- Ores: bauxite (aluminium), haematite (iron), cinnabar (mercury), galena (lead), zinc blende (zinc), malachite (copper).

- Extraction: highly reactive metals (Na, K, Al) by electrolysis. Medium (Zn, Fe, Pb) by reduction of oxides with carbon. Low reactivity (Hg) by heating the ore.

6. Chemical Reactions

- **Combination:** $2H_2 + O_2 \rightarrow 2H_2O$. **Decomposition:** $CaCO_3 \rightarrow CaO + CO_2$.
- **Displacement:** $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$. **Double displacement:** $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$ (white precipitate).
- **Oxidation** is gain of oxygen or loss of electrons. **Reduction** is loss of oxygen or gain of electrons. Both occur together (redox).
- Exothermic reactions release heat (combustion, respiration). Endothermic reactions absorb heat (photosynthesis).
- Law of conservation of mass (Lavoisier): mass is neither created nor destroyed in a chemical reaction.
- Photosynthesis: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (in presence of light and chlorophyll).

7. Carbon and its Compounds

- Carbon shows **catenation** (forms long chains) and has a valency of 4. Allotropes: diamond, graphite, fullerene (C-60), graphene.
- Hydrocarbons: alkanes (single bond, C_nH_{2n+2}, e.g. methane), alkenes (double bond, C_nH_{2n}, e.g. ethene), alkynes (triple bond, C_nH_{2n-2}, e.g. ethyne/acetylene).
- Functional groups: alcohol (–OH), aldehyde (–CHO), ketone (>C=O), carboxylic acid (–COOH).
- Ethanol (alcohol) is used in medicines, solvent and as fuel. Pure ethanol is called absolute alcohol. Methanol is poisonous. Ethanoic acid (acetic acid) is vinegar, and 5 to 8% solution is used in cooking.
- **Soap** is the sodium or potassium salt of long-chain fatty acids. Soap does not work well in hard water because it forms scum. Detergents work in hard water.
- Natural gas is mainly **methane** (CNG). LPG is mainly butane and propane. Biogas is mainly methane.
- Polymers: polythene, PVC, nylon, Teflon, bakelite are synthetic. Natural polymers: cellulose, starch, proteins, rubber.

8. Everyday Chemistry

Substance	Chemical name / content	Use
Baking soda	Sodium hydrogen carbonate	Cooking, antacid, fire extinguisher
Washing soda	Sodium carbonate	Cleaning, softening hard water
Bleaching powder	Calcium oxychloride	Disinfecting water, bleaching
Plaster of Paris	Calcium sulphate hemihydrate	Fracture supports, statues
Dry ice	Solid carbon dioxide	Cooling, ice-cream storage
Laughing gas	Nitrous oxide (N_2O)	Anaesthetic
Marsh gas	Methane	Fuel
Quicksilver	Mercury	Thermometers, barometers
Vitamin C	Ascorbic acid	Prevents scurvy
Rust	Hydrated iron(III) oxide	—
Vinegar	Acetic acid	Food preservative

- **Hard water** contains calcium and magnesium salts. Temporary hardness is removed by boiling; permanent hardness by washing soda.
- **Ozone (O_3)** in the stratosphere absorbs harmful UV rays. It is damaged by CFCs.
- **Greenhouse gases:** carbon dioxide, methane, nitrous oxide, water vapour, CFCs. **Acid rain** is caused by oxides of sulphur and nitrogen.
- **Fuel calorific values:** hydrogen has the highest calorific value among common fuels. Good fuels are cheap, burn easily and leave little residue.
- **Fertilisers:** urea (46% nitrogen) has the highest nitrogen content. NPK stand for nitrogen, phosphorus, potassium.
- Mole concept: 1 mole = 6.022×10^{23} particles (Avogadro number). Molar volume of a gas at STP = 22.4 litres.

KEY POINT: Remember the formulae and common names (baking soda, washing soda, bleaching powder, plaster of Paris) and the pH of common substances. These appear in NDA almost every year.